CSE 3421 · FINAL EXAM PREPARATION

Bilingual Study Guide · English + বাংলা

Five High-Priority Topics

1. Modern Software Engineering Today
2. REST vs SOAP
3. Responsible & Ethical AI
4. AI, LLMs & Prompt Engineering
5. Edge & IoT · Green Software Engineering

সহজ বাংলা ব্যাখ্যা, practical examples, comparison tables এবং পরীক্ষায় লেখার মতো English answers-সহ প্রস্তুত করা হয়েছে।

How to Use This Guide

Each topic includes an English definition, Bangla explanation, key points, examples, and an exam-ready answer.

৩ নম্বরের প্রশ্নে definition-এর সঙ্গে ২–৩টি ব্যাখ্যাসহ point লিখবে। বেশি নম্বরের প্রশ্নে একটি practical example যোগ করবে।

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TOPIC 01 · PAGES 3, 14–15

1. Modern Software Engineering Today

Definition

English: Modern software engineering is moving from mainly code-centric development toward data-driven, AI-assisted, automated, service-based, and cloud-enabled development.

বাংলা: আগে software development-এর মূল focus ছিল code লেখা। এখন data, AI, automation, cloud services, APIs এবং automated testing ও delivery-ও গুরুত্বপূর্ণ অংশ।

Traditional vs Modern Approach

Traditional	Modern
Mainly focuses on writing code	Uses data, AI and automation
Many tasks are manual	Many tasks can be automated
Deployment may be manual	CI/CD can automate delivery
Traditional infrastructure may be used	Cloud-native technologies can be used
Integration may be tightly coupled	APIs help services communicate

Cloud-Native Development

English: A modern approach to building and running applications that uses cloud computing to improve scalability, resilience, portability, and management.

বাংলা: Cloud-native পদ্ধতিতে cloud technology কাজে লাগিয়ে application চালানো, scale করা এবং manage করা সহজ হয়।

- **Containers:** Help applications run consistently across environments.
- **Kubernetes:** Orchestrates and manages container-based applications.
- **Serverless:** Hides much of the underlying server infrastructure from developers.

সহজভাবে: Container application-কে বিভিন্ন environment-এ চালাতে সাহায্য করে। Kubernetes অনেক container manage করে। Serverless-এ developer-কে server infrastructure-এর অনেক কিছু সরাসরি manage করতে হয় না।

API-Based Software

English: An API allows applications or services to communicate through defined requests and responses.

বাংলা: API হলো নির্দিষ্ট request ও response-এর মাধ্যমে আলাদা application বা service-কে যোগাযোগ করানোর মাধ্যম।

Example: A food delivery app uses a payment gateway API to send a payment request and receive the result.

উদাহরণ: Food delivery app payment করার জন্য payment gateway-এর API-তে request পাঠায় এবং payment result পায়।

CI/CD

Continuous Integration (CI): Developers frequently merge code into a central repository, triggering automated builds and tests.

বাংলা: Developer code repository-তে merge বা push করলে automated build ও test চালু হতে পারে।

Continuous Delivery (CD): Automates delivery so validated software can be released to target environments.

বাংলা: Test পাস করা software release বা নির্দিষ্ট environment-এ deliver করার প্রক্রিয়া automate করা হয়।

Tools: Jenkins, GitHub Actions, GitLab CI.

Exam-ready answer · 3 marks

Modern software systems are changing in four ways:

1. **AI and automation:** Reduce manual work and improve productivity.
2. **Cloud-native:** Improve scalability, portability, resilience, and management.
3. **API-based integration:** Allow applications and services to communicate and exchange data.
4. **CI/CD:** Automated builds, tests, and delivery help teams release software efficiently.

বাংলায় মনে রাখবে: AI ও automation, cloud-native, API-based integration এবং CI/CD—এই চারটি পরিবর্তন modern software development-কে দ্রুত ও দক্ষ করে।

2. REST vs SOAP

REST

English: REST (Representational State Transfer) is an architectural style for networked applications and web services.

বাংলা: REST হলো web service ও networked application-এর মধ্যে যোগাযোগের জন্য ব্যবহৃত architectural style।

REST's Five Constraints · C-S-C-L-U

- **Client–Server:** Client and server have separate responsibilities.
- **Stateless:** Each request carries the information needed to process it.
- **Cacheable:** Responses may be cached when appropriate.
- **Layered System:** Communication may pass through intermediary layers.
- **Uniform Interface:** Resources are accessed through consistent rules.

বাংলায়: Client–Server আলাদা কাজ করে; Stateless-এ প্রতিটি request প্রয়োজনীয় তথ্য বহন করে; Cacheable response সংরক্ষণ করতে পারে; Layered System-এ মাঝের layer থাকতে পারে; Uniform Interface-এ consistent নিয়মে resource access হয়।

SOAP

English: SOAP (Simple Object Access Protocol) is a protocol for exchanging structured information between applications.

বাংলা: SOAP হলো application-এর মধ্যে structured information আদান-প্রদানের একটি protocol।

REST	SOAP
Architectural style	Communication protocol
Lecture slide: low bandwidth	Lecture slide: high bandwidth
Supports multiple formats, including JSON and XML (per slide)	XML only (per slide)
Five constraints listed	No matching constraint list is given on the slide

Exam-ready answer · 3 marks

REST is an architectural style for web-service communication, whereas SOAP is a protocol for exchanging structured information between applications.

1. **Bandwidth:** The lecture slide describes REST as requiring low bandwidth and SOAP as requiring high bandwidth.
2. **Data format:** REST supports several formats such as JSON and XML; the slide describes SOAP as XML-only.
3. **Constraints:** REST follows Client–Server, Stateless, Cacheable, Layered System, and Uniform Interface.

3. Responsible & Ethical AI

English: Responsible and ethical AI means developing and using AI systems in ways that are fair, transparent, accountable, and aligned with ethical standards.

বাংলা: Responsible & Ethical AI হলো AI এমনভাবে তৈরি ও ব্যবহার করা, যাতে তা ন্যায্য, স্বচ্ছ, জবাবদিহিমূলক এবং নৈতিক মানদণ্ডের সঙ্গে সামঞ্জস্যপূর্ণ হয়।

Five Characteristics

Characteristic	English explanation	বাংলা ব্যাখ্যা	Example
1. Fairness	Avoid biases in AI models and decisions.	AI model এ পক্ষপাত এড়িয়ে চলা।	A hiring AI should not unfairly reject applicants.
2. Transparency	Make AI decisions understandable and explainable.	AI কেন সিদ্ধান্ত দিয়েছে, তা বোঝা ও ব্যাখ্যা করা যায়।	A loan system explains a rejection.
3. Accountability	Developers are responsible for AI outputs and impacts.	AI output ও প্রভাবের জন্য developer দায়ী থাকেন।	A team investigates and fixes harmful errors.
4. Privacy & Security	Protect sensitive data and maintain confidentiality.	সংবেদনশীল তথ্য সুরক্ষিত রাখা ও গোপনীয়তা বজায় রাখা।	A health app restricts access to patient records.
5. Sustainability	Design energy-efficient and socially responsible AI.	এমন energy-সাশ্রয়ী এবং সমাজের প্রতি দায়িত্বশীল AI তৈরি করা।	Reduce unnecessary AI computation.

Exam-ready answer · 3 marks

1. **Fairness:** Avoid biases in AI models and decisions.
2. **Transparency:** Make AI decisions understandable and explainable.
3. **Accountability:** Developers are responsible for AI outputs and impacts.
4. **Privacy and Security:** Protect sensitive data and maintain confidentiality.
5. **Sustainability:** Design energy-efficient and socially responsible AI systems.

Shortcut: F–T–A–P–S = Fairness, Transparency, Accountability, Privacy & Security, Sustainability.

TOPIC 04 · PAGES 17–19

4. AI, LLMs & Prompt Engineering

A. AI in Software Development

English: AI is integrated into software development to automate tasks, improve code quality, and accelerate development.

বাংলা: Software development-এ AI ব্যবহার করে কাজ automate করা, code quality উন্নত করা এবং development দ্রুত করা যায়।

- **Code generation:** Help write code from requirements.
- **Debugging:** Help identify and fix bugs.
- **Documentation:** Help create technical explanations.
- **Optimization:** Suggest improvements to code and workflows.

Examples: GitHub Copilot and ChatGPT. AI suggestions should be reviewed and tested before use.

B. LLMs in Software Engineering

English: Large Language Models (LLMs) understand and generate human-like text, including code. They can help generate code, fix bugs, suggest architectures, and automate repetitive tasks.

বাংলা: LLM এমন AI model, যা মানুষের মতো text ও code তৈরি করতে পারে। এটি code লেখা, bug fix-এর পরামর্শ, architecture suggestion এবং repetitive task-এ সাহায্য করে।

Examples: ChatGPT, Gemini, Claude. APIs can integrate AI features into software.

C. Prompt Engineering

English: Prompt engineering is the practice of crafting effective inputs for AI models to obtain accurate, relevant, and useful outputs.

বাংলা: AI-কে ভালো ও নির্দিষ্ট instruction দেওয়ার জন্য prompt পরিকল্পনা ও সাজানোর পদ্ধতিই Prompt Engineering।

Weak prompt	Better prompt
“Write code.”	“Write a Python function that checks whether a number is even or odd. Explain the code and include three test cases.”

Limitations

Hallucination: AI may produce incorrect or fabricated information and present it as true.

বাংলা: AI কখনো ভুল বা বানানো তথ্যকে সঠিক বলে উপস্থাপন করতে পারে।

Privacy: Sensitive information in prompts may expose private or confidential data.

বাংলা: Prompt-এ password, personal data বা confidential source code দিলে গোপনীয়তার ঝুঁকি তৈরি হতে পারে।

Examples: AI suggests a function that does not exist (hallucination); a developer pastes confidential code into an unapproved public tool (privacy risk).

Exam-ready answer · 3 marks

Prompt engineering is the practice of designing effective prompts to obtain accurate, relevant, and useful AI outputs.

Uses: Code generation, content creation, and data analysis.

Limitations: Hallucination means AI may generate incorrect information. Privacy means sensitive information in prompts may be exposed.

TOPIC 05 · PAGES 21–22

5. Edge & IoT; Green Software Engineering

A. Edge and IoT Computing

English: Edge and IoT computing process data locally, near the IoT devices that generate it, instead of sending all data to a central cloud.

বাংলা: IoT device থেকে তৈরি data দূরের cloud-এ পাঠানোর বদলে device-এর কাছাকাছি process করা হয়। এতে দ্রুত response ও local processing সম্ভব হয়।

Benefits

- **Low latency:** Nearby processing provides faster responses.
- **Offline capabilities:** Some local functions can continue without continuous internet access.
- **Real-time analytics:** Device data can be analyzed immediately.

Practical examples

- **Smart home:** A motion sensor triggers a light through a local controller.
- **Autonomous vehicle:** Processes sensor data quickly to detect obstacles.
- **Smart garbage management:** Bin sensors report fill levels for timely collection.

Exam-ready answer · 3 marks

Edge and IoT computing process data near IoT devices instead of sending all data to a central cloud.

Benefits: (1) Low latency, (2) offline capabilities through local processing, and (3) real-time analytics.

Examples: Smart homes, autonomous vehicles, and smart garbage management.

B. Green Software Engineering

English: Green Software Engineering focuses on minimizing environmental impact by improving energy efficiency and reducing carbon emissions throughout the software lifecycle.

বাংলা: Software-এর পুরো lifecycle-এ energy ব্যবহার ও carbon emissions কমিয়ে পরিবেশের ওপর প্রভাব সীমিত রাখার দিকে Green Software Engineering মনোযোগ দেয়।

Objectives

- **Optimize CPU and energy usage:** Reduce unnecessary computation and energy consumption.
- **Efficient cloud scaling:** Match cloud resources to workload and avoid unnecessary capacity.
- **Sustainability metrics:** Measure energy use and sustainability-related impacts.

Practical examples

- Optimize code to reduce CPU usage.
- Reduce unnecessary background tasks.
- Scale cloud resources down when demand is low.
- Measure application energy and resource consumption.

Exam-ready answer · 3 marks

Green Software Engineering minimizes environmental impact by improving energy efficiency and reducing carbon emissions throughout the software lifecycle.

Objectives: (1) Optimize CPU and energy usage, (2) efficient cloud scaling, and (3) sustainability metrics.

Example: Optimize an application to reduce CPU use and scale down unused cloud resources.

Final 1-Minute Revision

Topic	Remember these keywords
Modern Software Engineering	AI + automation · Cloud-native · APIs · CI/CD
REST vs SOAP	REST: style, five constraints, multiple formats · SOAP: protocol, XML (lecture slide)
Responsible & Ethical AI	F–T–A–P–S: Fairness, Transparency, Accountability, Privacy & Security, Sustainability
AI & LLMs	Code generation · Debugging · Documentation · ChatGPT/Gemini/Claude
Prompt Engineering	Clear prompts · Useful outputs · Hallucination · Privacy
Edge & IoT	Low latency · Offline capabilities · Real-time analytics
Green Software	CPU/energy optimization · Efficient cloud scaling · Sustainability metrics

Exam checklist

- Start with a one-sentence definition.
- Use the lecture's key terms.
- Explain each point in one simple sentence.
- Add a practical example when required.
- Use a table for comparison questions.

শেষ মুহূর্তে: definition, key points, এবং একটি example মনে রাখবে। Comparison question হলে table ব্যবহার করবে।

Source note: Core definitions and topic coverage follow the Current Trends in Software Engineering lecture material. Examples are included to make the concepts easier to understand.